



Improving Emotional Text-to-Speech in Low-Resource Languages via Reinforcement Learning

Hao-Chun Hsieh Pao-Hua Hsu Nien-Tse Wu

Institute of Artificial Intelligence Innovation, National Yang Ming Chiao Tung University

Abstract

We study emotional capability loss when an expressive TTS model trained on high-resource languages is adapted to Taiwanese. Building on IndexTTS2, the final system combines parameter-efficient adaptation, RL-based emotional realignment, and *VA-and-stress control*. The goal is to preserve the pretrained expressive prior while improving emotion intensity, lexical emphasis, and speech naturalness in Taiwanese.

Problem and Objectives

Low-resource adaptation often improves pronunciation but weakens the expressive behavior learned from high-resource speech. The final report frames this as an emotion preservation and alignment problem.

- **Emotional forgetting:** limited Taiwanese fine-tuning can make synthesized speech flatter and less controllable.
- **Prosodic control loss:** rhythm, pauses, local stress, and emotional intensity may degrade even when speech remains intelligible.
- **Cross-lingual alignment:** emotion cues do not transfer perfectly across Chinese, English, and Taiwanese phonology.
- **Objective:** recover controllable emotional expression through RL-based realignment and VA-conditioned lexical stress control.

Proposed Framework

The final method follows a two-stage IndexTTS2 pipeline with an annotation and phonemic text front-end placed between adaptation and evaluation.

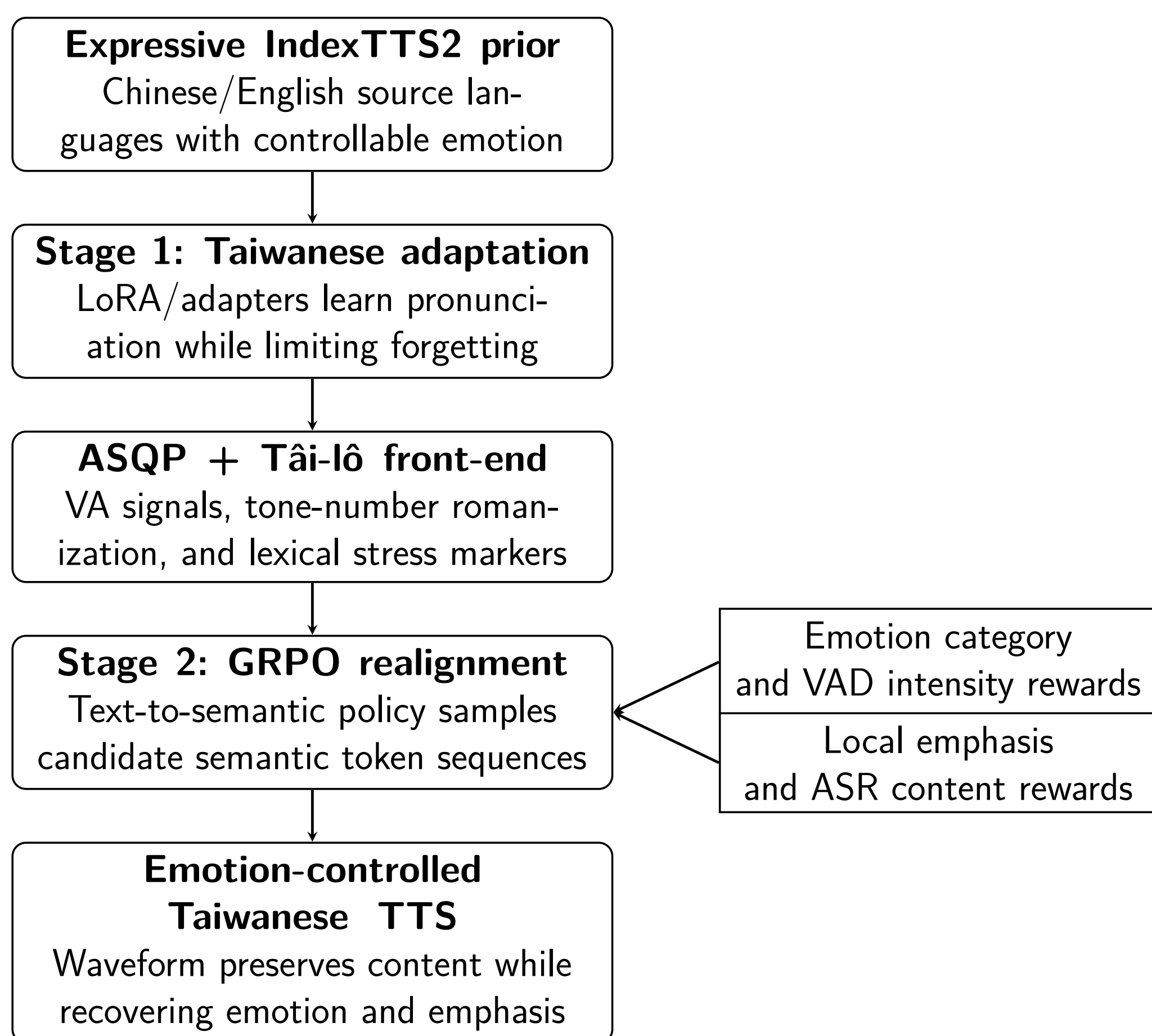


Figure 1. Final-report pipeline: low-resource adaptation followed by ASQP-conditioned RL realignment.

- **Stage 1:** LoRA or lightweight adapters learn Taiwanese pronunciation while limiting catastrophic parameter drift.
- **ASQP control:** Traditional Chinese aspect–sentiment annotations provide Valence–Arousal signals and sentiment-bearing spans.
- **Phonemic front-end:** Taiwanese text is rendered as Tâi-lô romanization with tone digits, while emphasis markers are kept visible to the policy.
- **Stage 2:** GRPO updates the autoregressive text-to-semantic policy using waveform-level rewards.

RL Formulation

Semantic speech generation is formulated as a fully observable MDP over the IndexTTS2 autoregressive decoder.

- **State:** decoder hidden state conditioned on input text, speaker embedding, duration, target emotion, and prior speech tokens.
- **Action:** the next discrete semantic speech token from policy $\pi_\theta(A_t|S_t)$.
- **Reward:** a sequence-level score computed after decoding the semantic tokens into a waveform.

$$R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}}$$

$$\max_{\theta} \mathbb{E}_{s,a \sim \pi_\theta} [R(s,a)] - \beta \text{KL}(\pi_\theta(\cdot|s) \parallel p_{\text{adapt}}(\cdot|s))$$

Data and Synthesis Setup

- **Training corpus:** TAT-MOE provides roughly 200 hours of native Taiwanese speech for low-resource adaptation.
- **Evaluation corpus:** TAT-Vol1 validation and test sets provide about 5 hours each; the synthesis study uses 2,752 utterances.
- **Emotion set:** each utterance is rendered with one of seven target categories: angry, happy, fearful, disgusted, sad, surprised, and neutral.
- **Controlled speaker:** all systems share one fixed Taiwanese reference speaker, producing 27:04:06 total synthesized audio with 44 failed generations across CosyVoice baselines.

Evaluation Metrics

- **SER:** emotion2vec recognition accuracy by target emotion and corpus-wide mean.
- **WER:** Taiwanese pinyin ASR evaluates intelligibility; the IndexTTS2 rows use the same tone-insensitive scoring pipeline.
- **SS:** ECAPA-TDNN speaker embeddings measure timbre retention against the fixed Taiwanese reference speaker.
- **MOS pilot:** 12 SARC members produced 600 ratings; CosyVoice2 reached 4.31 mean MOS, shifting the focus from quality to emotion recovery.

Architecture and Reward Design

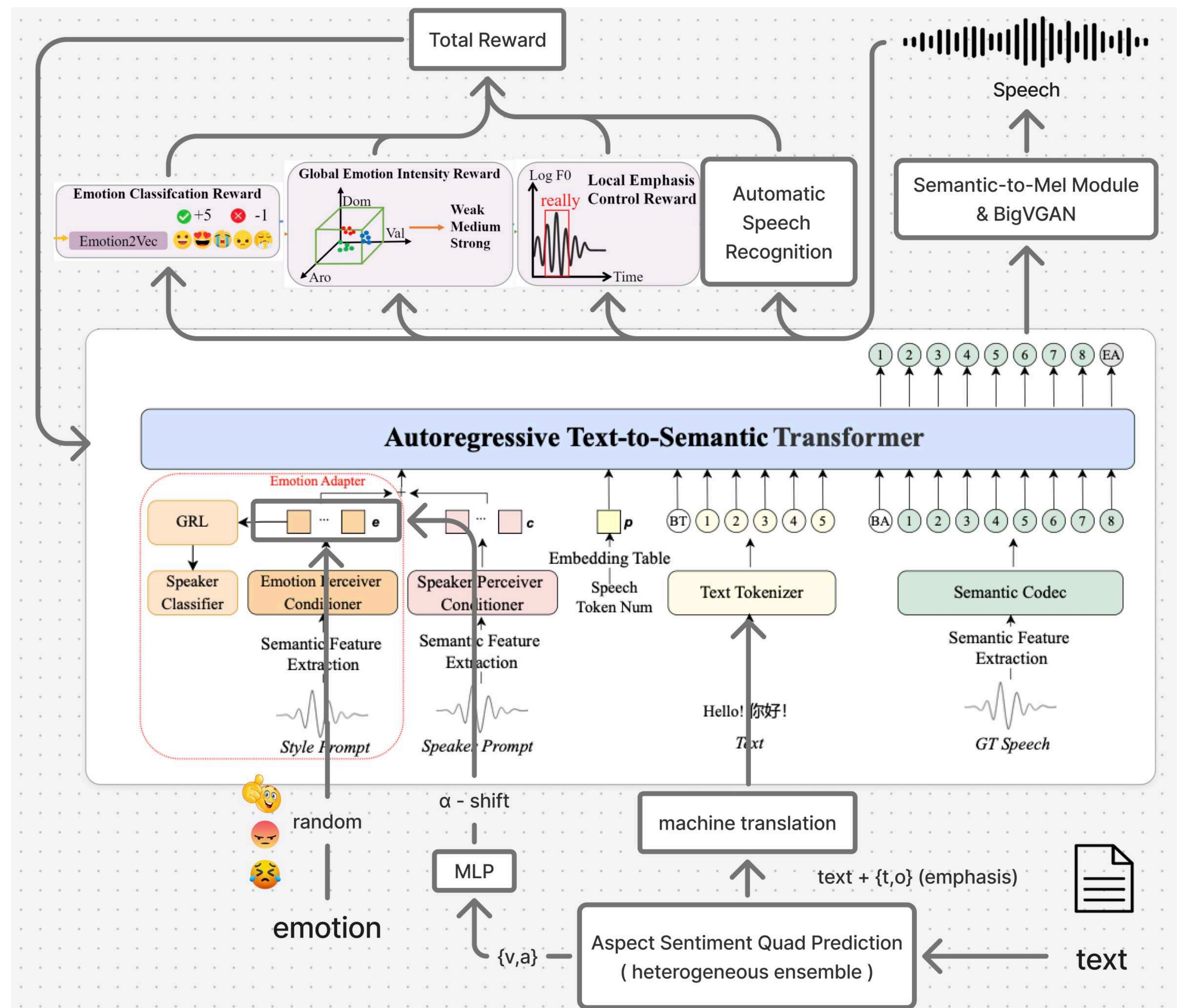


Figure 2. RL-enhanced IndexTTS2 architecture with waveform-level reward models.

Reward models evaluate decoded speech rather than only token likelihood.

- **Emotion category:** emotion2vec checks whether the waveform matches the requested emotion label.
- **Global intensity:** distance from a neutral centroid in Valence–Arousal–Dominance space encourages stronger affect.
- **Local emphasis:** pitch and energy patterns reward stressed words marked by the ASQP-derived control signal.
- **Content accuracy:** a Taiwanese pinyin ASR model evaluates whether emotional rendering preserves the input sentence.

Final Report Results

The final report evaluates emotion recognizability, intelligibility, and speaker fidelity across CosyVoice baselines and IndexTTS2 ablations.

Table 1. Corpus-wide final report metrics. SER and SS are higher-is-better; WER is lower-is-better.

Model	SER $\uparrow$	WER $\downarrow$	SS $\uparrow$
CosyVoice2	0.14	1.666 [†]	0.581
CosyVoice3	0.13	2.110 [†]	0.573
IndexTTS2-FT	0.27	0.705	0.456
IndexTTS2-FT+VA/stress	0.27	0.741	0.463
IndexTTS2-FT+RL	0.28	0.816	0.433
IndexTTS2-proposed	0.30	0.840	0.416

[†]CosyVoice WER is a tone-sensitive diagnostic score and is not directly comparable to the IndexTTS2 rows.

- **Emotion recognition:** all IndexTTS2 variants outperform CosyVoice2/3 in mean SER accuracy; the full proposed system is highest at 0.30.
- **VA-and-stress after RL:** control has little effect before RL (0.27 $\rightarrow$ 0.27 SER), but improves the RL checkpoint (0.28 $\rightarrow$ 0.30), driven by happy emotion accuracy (0.41 $\rightarrow$ 0.58).
- **Intelligibility trade-off:** WER rises from 0.705 to 0.840 from plain fine-tuning to the full system, a bounded cost for stronger emotional rendering.
- **Speaker fidelity:** stronger emotion control lowers speaker similarity, especially after RL, making timbre preservation the main next challenge.

Key Findings and Next Step

- IndexTTS2 moves to the opposite side of the baseline trade-off: better emotion recognition than CosyVoice2/3, but weaker speaker similarity.
- RL realignment reshapes the emotional profile, improving surprise and making VA-and-stress control more effective.
- Sad, fear, and disgust remain difficult under the current SER-based evaluation, suggesting that reward models and data balance need further refinement.
- Next stage: improve speaker similarity while retaining the emotion gains from RL realignment and VA-and-stress control.

Team and References

Team: Hao-Chun Hsieh – project direction, formulation, and methodology; Pao-Hua Hsu – audio generation, visualization, poster, and presentation; Nien-Tse Wu – implementation, dataset curation, evaluation, and tuning.

References: IndexTTS2; EMORL-TTS; emotion2vec; TAT-MOE/TAT-Vol1; CosyVoice2/3; FunASR; Whisper; ASQP/SemEval-2026 Task 3.